

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-0950

Donner 138 kV
March 05, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0950. The Interconnection Customer proposes a 210.3 MW solar photovoltaic facility interconnecting at the Donner 138 kV substation in the Delaware area. The request was received on February 26, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 269 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,566,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Donner to Ninemile Point 138 KV	Cady to Dodd Rd 345 KV	144.2	233	85.4
North Lebanon to Fairmont 138 KV	Cady Substation to Dodd Rd 345 KV Line	133.1	316	36.0
Willow Pass to Churchs Ferry 138 KV L	Dodd Road to Cady Substation 345 KV line	120.2	196	9.1
Ninemile Point to N Lebanon 138 KV Line 1	Dodd Rd to Cady 345 KV	118.8	365	36.0
N Lebanon to Ninemile Point 138 KV Cir. 2	Cady to Dodd Road 345 KV	118.8	365	36.0
Banta Carbona to Willow Pass 138 KV	Cady to Dodd Road 345 KV	118.7	259	12.4
Seneca to Mckee Run 138 KV Line	Cady to Dodd Rd 345 KV	113.4	398	8.4
Cady to Easton 2 345 KV line	Memphis Jct to Lammers Substation 138 KV	112.9	1,669	8.4
Mckee Run to Wheatland Sta 138 KV	Cady to Dodd Road 345 KV line	110.3	245	6.0
Banta Carbona Substation to Seneca 138 KV Line	Cady to Dodd Rd 345 KV Line	109.7	349	8.4
Fairmont to Bond-Elk Grove Florin 138 KV Circuit 1	Cady to Dodd Rd 345 KV	102.1	406	36.0
Bond-Elk Grove Florin Substation to Fairmont 138 KV Line 2	Cady to Dodd Road 345 KV	102.1	406	36.0
N Lebanon to Fairmont 138 KV #1	Dodd Road to Cady Substation 345 KV L	101.4	415	36.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Doner to Ninemile Point 138 KV line	Rebuild line with higher-capacity conductor	\$2,270,000	41% (\$923,000)
North Lebanon to Fairmont 138 KV	Rebuild line with higher-capacity conductor	\$3,713,000	12% (\$447,000)
Churchs Ferry to Willow Pass 138 KV line	Rebuild line with higher-capacity conductor	\$2,132,000	6% (\$117,000)
Ninemile Pt to North Lebanon 138 KV Cir. 1	Rebuild line with higher-capacity conductor	\$2,064,000	12% (\$248,000)
N Lebanon to Ninemile Point 138 KV Cir. 2	Rebuild line with higher-capacity conductor	\$1,891,000	12% (\$227,000)
Banta Carbona to Willow Pass 138 KV	Rebuild line with higher-capacity conductor	\$22,618,000	6% (\$1,236,000)
Mckee Run to Seneca 138 KV	Rebuild line with higher-capacity conductor	\$5,845,000	3% (\$180,000)
Cady to Easton 2 Substation 345 KV	Rebuild line with higher-capacity conductor	\$79,462,000	1% (\$556,000)
Mckee Run to Wheatland Sta 138 KV	Rebuild line with higher-capacity conductor	\$2,760,000	5% (\$137,000)
Seneca to Banta Carbona 138 KV	Reconductor line	\$999,000	3% (\$31,000)
Fairmont to Bond-Elk Grove Florin 138 KV	Reconductor line	\$1,101,000	12% (\$132,000)
Bond-Elk Grove Florin to Fairmont 138 KV Cir. 2	Reconductor line	\$1,082,000	12% (\$130,000)
N Lebanon to Fairmont 138 KV #1	Reconductor line	\$1,680,000	12% (\$202,000)
	Total allocated to project		\$4,566,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.